k-armed bandits

A stepping stone to the full RL problem

What you'll learn

- Introduce the k-armed bandit problem
- This is a simpler version of the full reinforcement learning problem
- Introduces many of the same concepts
- K-armed bandit algorithms underpin many real-world applications

My Instagram feed:

Inanimate objects under duress

Social feed recommendations

Social feed recommendations

Selected photo (action)	

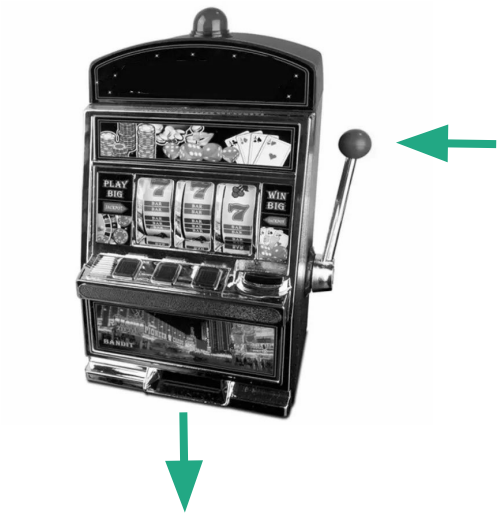
Social feed recommendations

Selected photo (action)				
Hover time (reward)	10.2s	11.5s	8.0s	20.7s

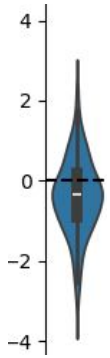
K-armed bandits

A one-armed bandit

- A one-armed bandit is a simple gambling machine with a single lever
- You have one action - pulling the lever - which you can repeat over time
- At each time step, we take our only available action (pull the lever)
- We observe a reward (or loss) at that time step
- Reward sampled from a (typically unknown) random distribution
- Over time, we aim to understand the machine's behavior through repeated interactions



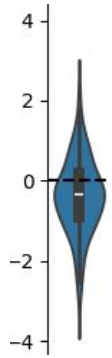
One-armed bandit // definitions



t	0	1	2	3	4	5
R_t	-1.2	0.3	-0.8	0.1	-1.5	0.6

One-armed bandit // definitions

- R_t – the random variable representing the reward at timestep t.
- $\mathcal{R}$ – the set of all possible values of r, where r is a single realization of R_t.
- $p(r)$ is the probability distribution of the rewards.
- $r \sim p(r)$ expresses that a single value of R_t is drawn from its distribution



The set $\mathcal{R}$ can be:

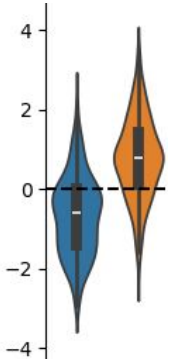
- A subset of natural numbers (for discrete rewards): $\mathcal{R} \subseteq \mathbb{N}$
- A subset of real numbers (for continuous rewards): $\mathcal{R} \subseteq \mathbb{R}$

More specifically, we can express common supports as:

- Discrete, finite set: $\mathcal{R} = \{r_1, r_2, ..., r_n\}$
- Continuous, bounded interval: $\mathcal{R} = [a, b]$, where $a, b \in \mathbb{R}$ and $a < b$
- Continuous, semi-bounded interval: $\mathcal{R} = [0, \infty)$

t	0	1	2	3	4	5
R_t	-1.2	0.3	-0.8	0.1	-1.5	0.6

A two-armed bandit



t	0	1	2	3	4	5
$R_t(A_t = 0)$	0.1		-0.8	0.1		0.6
$R_t(A_t = 1)$		-0.5			1.8	

A two-armed bandit

- Two-armed bandits introduce a choice between actions: $A_t \in \mathcal{A} = \{0, 1\}$
  - Action 0: Pull the lever on machine 1
  - Action 1: Pull the lever on machine 2

- The reward R_t at time t depends on the chosen action A_t

- Each machine has its own reward distribution:
  - $R_t(A_t = 0) \sim p_0(r|A_t = 0)$ for machine 0
  - $R_t(A_t = 1) \sim p_1(r|A_t = 1)$ for machine 1

- The challenge: Learn which machine offers better rewards over time

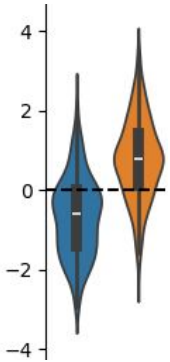
$$A_t \in \mathcal{A} = \{0, 1\}$$

$$R_t(A_t = 0)$$

$$R_t(A_t = 1)$$

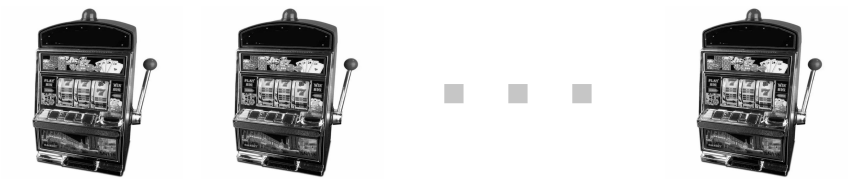
$$R_t(A_t = 0) \sim p_0(r|A_t = 0)$$

$$R_t(A_t = 1) \sim p_1(r|A_t = 1)$$

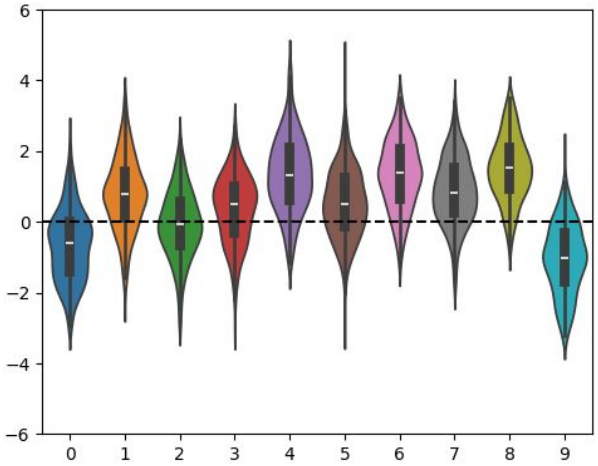


t	0	1	2	3	4	5
$R_t(A_t = 0)$	0.1		-0.8	0.1		0.6
$R_t(A_t = 1)$		-0.5			1.8	

A k-armed bandit



- K-armed bandits generalise the concept to k different actions (or "arms")
  - Each arm represents a distinct strategy or choice
  - Actions: A_t ∈ A = {0, 1, ..., a, ...(k-1)}
  - Typically, k is fixed and known, but can be large
- Each action yields a stochastic reward R_t(A_t = a)
  - Rewards come from unknown, action-specific distributions
- The true value of an action, q*(a), is its expected reward
  - This value is unknown and must be estimated through experience
- Challenge: Balance exploration (trying new arms) with exploitation (choosing the best known arm)
- Goal: Maximise cumulative reward over time by learning the best action(s)



$$A_t \in \mathcal{A} = \{0, 1, ..., a, ...(k-1)\}$$

$$R_t(A_t = a)$$

$$q_*(a) \doteq \mathbb{E}[R_t(A_t = a)]$$

Summary & wrap-up

- **Introduced the k-armed bandit problem**
  - One-armed bandit to introduce the reward random variable R_t
  - Two-armed bandit to introduce the action random variable A_t
  - K-armed bandits to generalise

- **A simpler version of the full RL problem but introduces similar concepts**
  - Rewards and actions
  - Typically unknown reward distributions
  - True value functions as expected rewards from actions

- **Got an initial feel for the explore-exploit trade-off**

- **K-armed bandit algorithms underpin many real-world applications**
  - Recommender systems
  - A/B testing
  - Clinical trials
  - Dynamic pricing